ASEAN
GEOAI FUSION 2026

Module 4:
Foundational Knowledge: Foundation in GEOAI (FD1004)

AGAIF 2026 Boot Camp 1
Level: Foundation of GeoAI

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From Connectivity Gaps to Evidence-Based Decisions

The learning journey across Sessions 4–6

SESSION 4

UNDERSERVED COMMUNITIES' IDENTIFICATION

UNDERSTAND & IDENTIFY

Who may be underserved — and why?

Combine telecommunications conditions with population, community and spatial context.

Telecom Network · Population · Communities · Public Services

Potential underserved communities

SESSION 5

CASE STUDY – ASEAN TELECOMMUNICATIONS CHALLENGE

COMPARE & REASON

When connectivity needs compete, what evidence should guide priority?

Examine ASEAN telecommunications challenges, evidence layers and competing planning considerations.

Impact · Urgency · Feasibility · Public Value

Transparent priority reasoning

SESSION 6

HANDS-ON

APPLY & RECOMMEND

How can spatial evidence be transformed into a defensible recommendation?

Apply the guided spatial screening workflow, interpret the evidence and extend it into priority and intervention recommendations.

Jupyter Workflow · Spatial Screening · Candidate Areas · Recommendation

Evidence-based decision support

Why This Module Matters

Connectivity inequality is not only a network problem. It is a spatial problem involving people, place, public services and planning priorities.

GeoAI helps connect these evidence layers so that connectivity decisions can be explained and defended.

IDENTIFY → **COMPARE** → **APPLY** → **RECOMMEND**

Learning progression: Understand the problem → Reason with evidence → Apply the workflow → Support a decision

Session 5:

Case Study – ASEAN Telecommunications Challenge

Case Study – ASEAN Telecommunications Challenge

Objectives — From Challenge Understanding to Evidence-Based Prioritisation

SESSION OBJECTIVES

01 Understand the **real-world telecommunications challenges** faced across diverse **ASEAN regions** with competing priorities.

02 Explore and evaluate **key data layers** that inform connectivity decisions and understand what each layer contributes.

03 Develop **multi-criteria reasoning** to identify and rank underserved areas based on impact, vulnerability, and feasibility.

04 Apply **prioritisation logic** to compare areas, justify why some areas should be prioritised over others, and recommend suitable interventions.

05 Prepare for the **hands-on challenge** by understanding the expected outputs and evaluation approach.

By the end of this session

> Participants should be able to think like connectivity planners by combining data, reasoning, and priorities to make **defensible intervention decisions.**

Key takeaway:
This session moves from challenge exploration to building the reasoning needed for evidence-based intervention decisions.

Challenge understanding → Data layer exploration → Priority reasoning → Intervention recommendation → Expected outputs

Case Study – ASEAN Telecommunications Challenge

From Underserved Mapping to ASEAN Connectivity Decisions

SESSION CONTEXT

01 Session 4 established the GeoAI logic for identifying **underserved communities** using coverage, population, vulnerability, and feasibility evidence.

02 Session 5 reframes this into a **regional decision problem** where resources are limited and not all areas can be served at once.

03 Teams must compare **diverse ASEAN scenarios**: urban congestion, rural interiors, islands and coastal areas, border communities, and education or health priority zones.

04 The task is not only to map underserved areas, but to justify where investment should go first using **evidence-based prioritisation**.

By the end of this session

> Participants should be able to move from descriptive mapping to **defensible connectivity decisions** by combining context, evidence, and prioritisation logic.

Key takeaway:
This session transforms GeoAI concepts into a regional challenge briefing for evidence-based connectivity planning.

Session 4 logic → Regional scenarios → Evidence layers → Priority reasoning → Decision outputs

Case Study – ASEAN Telecommunications Challenge

Problem Scenario: Where Should Connectivity Investment Go First?

DECISION PROBLEM

Region Type	Example Challenge
Urban centres	congestion, service quality degradation
Rural interiors	sparse infrastructure, difficult terrain
Islands / coastal areas	backhaul limitations, disaster vulnerability
Border communities	uneven investment, administrative complexity
Education / health zones	need reliable service for schools, clinics, telemedicine
Economic corridors	high demand, infrastructure pressure

Key question:
Which areas should be prioritised first, and why?

By the end of this scenario

> Participants should recognise that **limited resources**, **competing needs**, and **regional diversity** mean not all underserved areas can be prioritised at once.

Key takeaway:
Effective connectivity planning is a prioritisation problem, not just a coverage-mapping exercise.

Limited resources · Competing needs · Regional diversity · Priority logic · Decision focus

Case Study – ASEAN Telecommunications Challenge

Scenario: Rural Interiors







RURAL INTERIORS

Country / Region: ASEAN / Asia-Pacific

What this case highlights: Rural connectivity remains significantly behind urban connectivity across ASEAN and the wider Asia-Pacific region.

Key evidence:

01	02	03
ASEAN Digital Masterplan 2025 notes that many ASEAN Member States still have rural areas where connectivity is poor or non-existent.	In 2019, only 37% of rural households had Internet access, compared with 70.4% of urban households in Asia-Pacific.	In 2024, 5G coverage reached 26% of urban populations but only 41% of rural populations in Asia-Pacific.

Planning gap / decision challenge: Lower population density, difficult terrain, and weaker commercial return often slow rural broadband expansion.

Why it matters: Connectivity prioritisation must balance rural inclusion, public value, feasibility, and investment efficiency.

Possible intervention directions: Shared infrastructure, fibre backhaul extension, universal service support, public Wi-Fi, and alternative access for hard-to-reach communities.

Key message

> Rural interiors illustrate why underserved connectivity is a prioritisation problem, not only a coverage problem.

Decision cue:
Should rural inclusion be prioritised even when deployment cost is high?

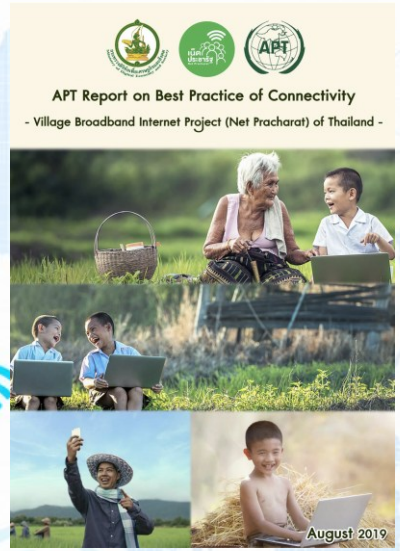
Key takeaway:
Rural interiors show that prioritisation must balance inclusion, feasibility, and investment efficiency.

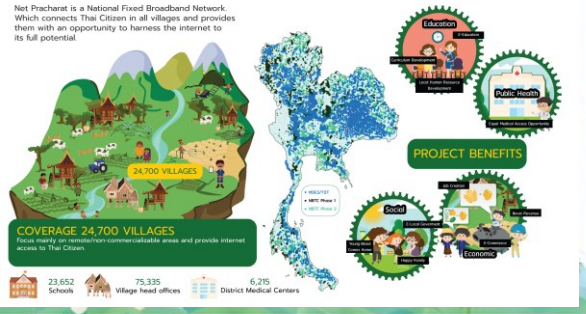
Evidence base → Scenario focus → Regional perspective → Priority reasoning → Decision outputs

Sources: ASEAN Digital Masterplan 2025 (ASEAN Secretariat, 2021) | Measuring Digital Development: Facts and Figures 2025 / Asia-Pacific data (ITU) | Digital Trends in Asia and the Pacific 2021 (ITU)

Case Study – ASEAN Telecommunications Challenge

Scenario: Rural Village Broadband





RURAL VILLAGE BROADBAND

Country / Region: Thailand

What this case highlights: Targeted public broadband intervention can extend fibre and free Wi-Fi access to previously underserved rural villages.

Key evidence:

01 Thailand's Net Pracharat project deployed fibre to 24,700 rural villages.

02 Each target village received one free public Wi-Fi hotspot at 30/10 Mbps.

03 The initiative reached millions of users, showing the scale of village-level broadband inclusion.

Planning gap / decision challenge: Low-density and remote rural villages may remain underserved because rollout costs are high and market-led expansion may not prioritise them.

Why it matters: This case shows that public investment can directly target underserved villages and improve connectivity inclusion beyond commercially attractive areas.

Possible intervention directions: Fibre extension, public Wi-Fi, universal service support, community access points, and rural digital literacy enablement.

Key message

"Rural village broadband shows that underserved connectivity can be reduced through targeted, village-level public broadband programmes."

Decision cue:
Should public broadband programmes prioritise rural villages that are not commercially attractive to operators?

Key takeaway:
Rural village broadband highlights the value of targeted public intervention for last-mile inclusion.

Evidence base → Village access → Public investtment → Priority reasoning → Decision outputs

Sources: Village Broadband Internet Project (Net Pracharat) of Thailand, Asia-Pacific Telecommunity (2019) | Net Pracharat: Thailand Project Description (EN), MDES / TOT (2019)

Case Study – ASEAN Telecommunications Challenge

Scenario: Islands / Remote Public Facilities



SpaceX's Falcon-9 rocket carrying the Republic of Indonesia Satellite-1 (SATRIA-1) lifts off from Cape Canaveral Space Launch Complex (SLC 40) in Florida, the United States, Sunday (June 18, 2023). (ANTARA/Livia Kristianti/aa/nbl).

ISLANDS / REMOTE PUBLIC FACILITIES

Country / Region: Indonesia

What this case highlights: Geographically fragmented and remote areas may require satellite-based connectivity to serve schools, clinics, local government offices, and other public facilities in 3T regions.

Key evidence:

01 Indonesia's SATRIA-1 satellite will deliver broadband to 50,000 public facility points in remote 3T regions.

02 The service is designed to provide approximately 4 Mbps connectivity for each supported public facility point.

03 SATRIA-1 is designed to serve up to 149,400 public service points, with over 30,000 locations already connected by early 2024.

Planning gap / decision challenge: Fragmented islands, difficult terrain, and backhaul limitations make fibre-first rollout impractical in many remote locations.

Why it matters: This case shows that connectivity prioritisation must consider geography, public service needs, and technology suitability—not only terrestrial coverage expansion.

Possible intervention directions: Satellite connectivity, hybrid backhaul, resilient public-facility links, and complementary terrestrial rollout where feasible.

Key message

"Remote public facilities show that underserved connectivity is also a technology-choice problem, not only a coverage problem."

Decision cue:
Should isolated schools and clinics be prioritised even if satellite delivery is costlier than mainland rollout?

Key takeaway:
Remote public facilities highlight the need to match connectivity solutions to geography, service needs, and technology feasibility.

Evidence base → Geographic constraints → Public facility focus → Technology suitability → Decision outputs

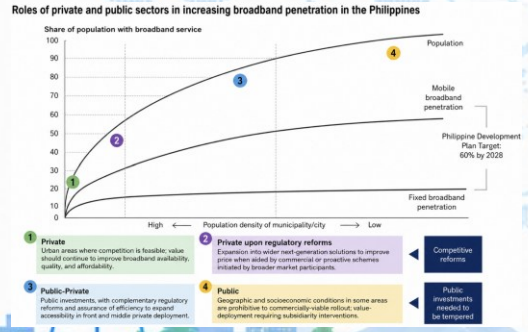
Sources: SATRIA-1 launch to improve internet access across Indonesia, Ministry of Communication and Informatics / ACN Newswire (2023) | Connecting the Nation from Space (SATRIA-1), PT Penjaminan Infrastruktur Indonesia / IIGF (2025)

Case Study – ASEAN Telecommunications Challenge

Scenario: Geographically Isolated & Disadvantaged Areas

"Improved internet access will help all Filipinos, especially those in remote areas of Mindanao, gain better education, healthcare, social protection, and government services," said **Zafer Mustafaoğlu, World Bank Country Director for the Philippines, Malaysia, and Brunei Darussalam**. *"This project fosters a brighter future for lagging regions by connecting everyone and ensuring that the poorest and most vulnerable Filipinos can benefit from the power of technology."*

To ensure resilience of these investments, the project will incorporate climate data, risks, and resilience measures into the design of the infrastructure. These measures include using climate-resilient materials and conduit systems for underground cables, designing overground infrastructure to withstand strong winds and floods, and incorporating innovative engineering solutions to adapt to climate-related risks, among others.



GEOGRAPHICALLY ISOLATED & DISADVANTAGED AREAS

Country / Region: Philippines

What this case highlights: Geographically isolated and disadvantaged areas may remain underserved because of remoteness, uneven infrastructure, public-service access needs, and affordability barriers.

Key evidence:

01 The Philippine DICT plans to connect 11,964 public schools in GIDA areas to the national Free Wi-Fi program by end-2025.

02 The World Bank's Philippines Digital Infrastructure Project supports national fiber backbone and connectivity to rural schools and health facilities, benefiting over 20 million Filipinos.

03 About 25% of Filipinos remain offline, many in geographically isolated and disadvantaged areas where reliable, affordable internet remains out of reach.

Planning gap / decision challenge: Archipelagic geography, remote communities, unequal infrastructure, and public access constraints make last-mile connectivity difficult in GIDA locations.

Why it matters: This case shows that prioritisation should consider who is affected, which services are at risk, and how public access can reduce exclusion.

Possible intervention directions: Free public Wi-Fi, school and clinic connectivity, fiber backbone expansion, middle-mile links, and targeted support for underserved island communities.

Key message

"Geographically isolated and disadvantaged areas show that underserved connectivity is also a public-service and inclusion problem, not only a network problem."

Decision cue:

Should schools, clinics, and isolated communities be prioritised even when reaching them requires higher public investment?

Key takeaway:
GIDA areas highlight the need to prioritise connectivity using inclusion, public service impact, and access feasibility.

Evidence base → Access inequality → Public service focus → Inclusion priority → Decision outputs

Sources: DICT to link 12K GIDA schools to Free WiFi program by year-end, Philippine News Agency (2025) | Unlocking the Philippines' Digital Transformation..., World Bank (2025) | A New Internet Era for the Philippines, Internet Society (2025) | Free Wifi for All in GIDA (Western Mindanao), ITU WSIS (2022)

Case Study – ASEAN Telecommunications Challenge

Scenario: Remote, Mountainous, Ethnic Minority & Border Areas

The government has been actively developing and adopting digital technology in its governance and public services. Over the past decade, the government has introduced several e-governance and public e-service programmes, including online tax, water, and electricity payments; electronic visa (e-Visa); online passport application; and intellectual property online application (IP e-Filing) (Mukherji et al., 2022). To support these initiatives, the country has continued to expand its telecommunications and high-speed internet infrastructure. In 2019, Lao PDR completed the installation of 90,258 kilometres of fibre optic cable, providing internet coverage to every district and province. Additionally, 7,882 cellular base stations were installed, covering 95% of the country's villages (Mukherjiet al., 2022). With these infrastructures in place, 3G and 4G networks now cover 82% of all districts and provinces and 55% of villages (Mukherji, P., Chatterjee, R., Sen, A., Kapoor, H., and Sambhar, A. 2022). In 2020, the country conducted a 5G network trial in select areas of its capital, Vientiane (Lao New Agency, 2020), and on 10 January 2024, during Lao Digital Week, it unveiled its first 5G internet service, operated by Lao Telecom (Laotian Times, 2024).


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HOME > SOCIETY

MIC pushes for expansion of broadband access to remote areas

November 14, 2024 - 10:16

0:00 / 0:00

Between 2021 and September 2024, about 3,310 villages were identified as lacking mobile broadband coverage. Of these, 2,141 villages are situated in areas supported by the programme, which focuses on regions with particularly difficult socio-economic conditions.

REMOTE, MOUNTAINOUS, ETHNIC MINORITY & BORDER AREAS

Country / Region: Vietnam / Laos

What this case highlights: Connectivity gaps in difficult terrain and vulnerable communities are shaped by remoteness, socio-economic conditions, and border, island, and mountainous contexts.

Key evidence:

01 Vietnam identified about 3,310 villages lacking mobile broadband between 2021 and September 2024.

02 The programme targets disadvantaged areas, border regions, islands, ethnic minority communities, and mountainous areas.

03 In Lao PDR, 90,258 km of fibre and 7,882 cell towers covered every district and 95% of villages, yet only 55% of villages had 4G coverage.

Planning gap / decision challenge: Terrain difficulty, remoteness, and socio-economic vulnerability make rollout slower and more uneven across hard-to-reach communities.

Why it matters: This case shows that prioritisation should consider who remains excluded, how difficult service delivery is, and whether vulnerable border and ethnic-minority communities need stronger public support.

Possible intervention directions: Shared infrastructure, targeted universal service support, mountain and border backhaul solutions, hybrid terrestrial-satellite coverage, and inclusive rollout for hard-to-reach communities.

Key message

"Remote and vulnerable communities show that underserved connectivity is shaped by geography, vulnerability, and border conditions—not only by market demand."

Decision cue:
Should difficult border and mountainous communities be prioritised even when rollout is slower and more costly?

Key takeaway:
Hard-to-reach and vulnerable communities require priority decisions that balance inclusion, terrain difficulty, and service feasibility.

Evidence base → Terrain challenge → Vulnerability focus → Priority reasoning → Decision outputs

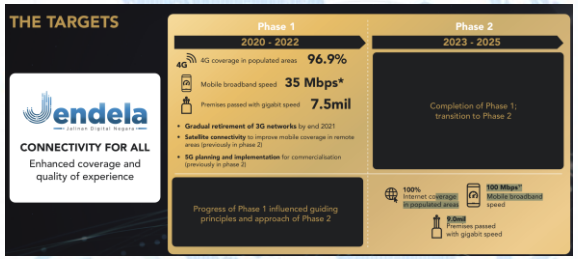
Sources: MIC pushes for expansion of broadband access to remote areas, Viet Nam News (2024) | ASEAN Digital Community 2045 – Lao PDR, ERIA (2022)

Case Study – ASEAN Telecommunications Challenge

Scenario: National Investment / Coverage Planning



Project	Plan (2020-2022)	Completed (Q2 2022)
New 4G Tower	3,576	1,479
Upgrade	37,095	34,136
Fiberisation	2,527,184	2,239,324
BWA	839	839
PoP	677	69
5G	4,018	1,471



Palapa ring project to boost ICT development in Indonesia

Indonesia | ICT

ANALYSIS

In an effort to bridge the geographical divide in digital services and provide high-speed internet across the country, the Widodo administration began work in 2016 on the Palapa Ring project. The priority infrastructure project comprises a 35,000-km fibre-optic cable network intended to provide access to 4G services throughout the archipelago by 2019. Some 34 provinces and more than 500 regencies will benefit from the project, especially those where it has not been viable for telecoms operators to lay cables, and increased connectivity should promote development within and between the country's outlying islands.

The $1.3bn project is funded primarily by loans from state institutions, including Indonesia Infrastructure Finance and state-owned finance company Sarana Multi Infrastruktur. Of the project's three sections, the west and central portions have already been completed, while the administration is hoping to complete the eastern section by the middle of 2019.

NATIONAL INVESTMENT / COVERAGE PLANNING

Country / Region: Malaysia / Indonesia

What this case highlights: National broadband plans can accelerate large-scale coverage expansion, but they still require evidence-based local prioritisation to decide where intervention should happen first.

Key evidence:

01 Indonesia's Palapa Ring built a 35,000 km fibre network across 34 provinces, supporting broader access and village connectivity.

02 Malaysia's JENDELA Phase 1 achieved 7.74 million premises passed, 96.9% 4G coverage of populated areas, and average broadband speeds of 116.03 Mbps.

03 JENDELA set national targets of 96.9% 4G coverage at 35 Mbps by 2022 and 100% coverage at 100 Mbps by end-2025.

Planning gap / decision challenge: National targets improve overall coverage, but they do not automatically reveal which underserved localities, technologies, and investment sequences should be prioritised first.

Why it matters: This case shows that macro-level broadband planning must still be translated into local, evidence-based intervention priorities for underserved communities.

Possible intervention directions: Priority mapping, targeted last-mile rollout, shared infrastructure, fibre backbone extension, mobile coverage enhancement, and localised universal service planning.

Key message

"National investment planning shows that underserved connectivity is also a sequencing and prioritisation problem, not only a coverage target problem."

Decision cue:
Should national broadband investment prioritise broad coverage targets first, or underserved localities with the highest social impact?

Key takeaway:
National broadband plans need local prioritisation logic to translate coverage targets into equitable intervention decisions.

Evidence base → National planning → Coverage targets → Priority reasoning → Decision outputs

Sources: Palapa Ring project to boost ICT development in Indonesia, Oxford Business Group (2019) | Contractors face action for Jendela delay..., The Star / Bernama (2024) | Jalinan Digital Negara (JENDELA) Plan, ITU WSIS Prizes (2023)

Case Study – ASEAN Telecommunications Challenge

Activity 1: Priority Vote

If resources are limited, which ASEAN connectivity scenario should be prioritised first?

VOTING OPTIONS

- **A** Rural Interiors 
- **B** Rural Village Broadband 
- **C** Islands / Remote Public Facilities 
- **D** Geographically Isolated & Disadvantaged Areas 
- **E** Remote, Mountainous, Ethnic Minority & Border Areas 
- **F** National Investment / Coverage Planning 

HOW TO VOTE

- Choose ONE scenario.
- Discuss in pairs or small groups.
- Justify your choice in 30 seconds.
- Use evidence from the previous case slides.

VOTING CRITERIA

Impact	Who and how many are affected?
Urgency	What happens if the gap remains unresolved?
Feasibility	What intervention is realistically possible?
Public value	Does it support inclusion, education, health, or national development?

Key takeaway:
Prioritisation is a decision-making exercise, not only a mapping exercise.

Evidence-based discussion → Priority reasoning → Trade-off thinking → Group justification → Decision readiness

Case Study – ASEAN Telecommunications Challenge

Evidence Layer Options

What evidence layers do we need to support a strong connectivity decision?

1. Network coverage / speed



Identifies where connectivity quality is weak.

2. Population grid



Shows who and how many are affected.

3. Schools / clinics / public facilities



Highlights which essential services are exposed.

4. Roads / settlements



Shows where communities are located and how accessible they are.

5. Towers / fibre / backhaul



Indicates how close the area is to existing infrastructure.

6. Terrain / slope / land cover



Represents physical difficulty of deployment.

7. Socio-economic indicators



Captures vulnerability and exclusion risk.

8. Administrative boundaries



Supports reporting and policy planning units.

Why this matters

- One dataset is not enough.
- A strong decision combines technical, social, infrastructure, and policy evidence.
- GeoAI turns multiple layers into explainable connectivity intelligence.

Connectivity Priority Evidence Base

Impact | Urgency | Feasibility | Public value

Key takeaway:
A strong connectivity decision is built from multiple evidence layers, not from a single map.

Data layers → Evidence fusion → Context understanding → Decision support → GeoAI neaciness

Case Study – ASEAN Telecommunications Challenge

Activity 2: Choose Your 4 Data Layers

Your team can only use 4 evidence layers. Which 4 will you choose to support your priority decision?

1. Network coverage / speed



2. Population grid



3. Schools / clinics / public facilities



4. Roads / settlements



5. Towers / fibre / backhaul



6. Terrain / slope / land cover



7. Socio-economic indicators



8. Administrative boundaries



Different data choices may lead to different priority decisions.

Team selection board

Our selected 4 layers

1.
2.
3.
4.

Our reason

Because our priority decision focuses on:

- ☐ Impact
- ☐ Urgency
- ☐ Feasibility
- ☐ Public value

Prepare a 30-second justification.

Key takeaway:
Dataset choice shapes the decision.

Data layers → Evidence fusion → Context understanding → Decision scoring → GeoAI summary

Case Study – ASEAN Telecommunications Challenge

Activity 3: Build the Priority Logic

Two areas are underserved. Which area should rank higher for intervention – and why?

Area X

Severe network gap, lower exposure

Network quality:	Very weak
Population affected:	Low
Schools / clinics:	None
Distance to tower / fibre:	Very far
Terrain difficulty:	High
Socio-economic vulnerability:	Medium
Intervention feasibility:	Low

Area X vs Area Y

Which area should be prioritised first?

Use these criteria

Impact	Who and how many are affected?
Urgency	What happens if the gap remains unresolved?
Feasibility	What intervention is realistically possible?
Public value	Does it support education, health, inclusion, or national development?

Prepare a 30-second justification.

Area Y

Moderate network gap, higher public impact

Network quality:	Moderate weakness
Population affected:	High
Schools / clinics:	3 schools + 1 clinic
Distance to tower / fibre:	Moderate
Terrain difficulty:	Medium
Socio-economic vulnerability:	High
Intervention feasibility:	Medium

Priority logic

Network deficit ····· People affected ··· Public value ····· Vulnerability ····· Feasibility

Key takeaway:
A strong priority decision balances network deficit, people affected, public value, vulnerability, and feasibility.

Collaborate effectively ····· Use data intelligently ····· Think critically & logically ····· Communicate clearly ····· Act for inclusion ad impact

From Case Study Reasoning to Hands-On Application

What the analysis should ultimately produce

EXPECTED OUTPUTS

01	**Underserved Priority Map**	Map showing high-priority underserved areas
02	**Top Priority Areas**	Ranked list of selected grids, districts, villages, or communities
03	**Evidence-Based Justification**	Explanation using coverage, population, vulnerability, facilities, and feasibility
04	**Recommended Intervention Type**	Tower, fibre, backhaul upgrade, satellite, public Wi-Fi, or shared infrastructure
05	**Policy / Investment Note**	How the recommendation supports planning, funding, or public service delivery
06	**Uncertainty Note**	Data gaps, limitations, confidence level, or assumptions

From analysis to decision-ready output

Map output → Priority ranking → Justification → Intervention → Decision note

Key message:
What should a strong team produce?

Hands-On Transition

"These outputs provide the decision-ready structure that participants will apply in the hands-on challenge."

Key takeaway:
A strong case study output translates spatial analysis into clear, evidence-based next steps for action.

Actionable outputs → Transparent reasoning → Policy relevance → Intervention readiness → Hands-on session

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THANK YOU

Bootcamp and Hackathon for ASEAN Geospatial Innovation 2026